

# THERMAL POWER — FULL MIND MAP

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## Topics in this chapter:

1. Singrauli NTPC — UP-MP Border, NOT Chhattisgarh
2. Idukki — Hydro (Not Thermal), Match-Set Verification
3. Thermal Plant-State Match (Ukai, Patratu, Pench, Dabhol)
4. Uran — Gas-Based Plant in Maharashtra
5. Coal — India's Largest Energy Source (52.89%)
6. Thermal Power — Highest Share in Generation
7. Energy Source Ranking (Disputed — No Fixed Sequence)
8. 2019 Installed Capacity Order (Water > Gas > Nuclear > Diesel)
9. Farakka — NTPC Superthermal Plant in West Bengal
10. Map-Based — Srisailem is Hydro, Not Thermal
11. Coal-Based Plant Match (Kahalgau-Bihar, Khargone-MP)
12. Neyveli — Tertiary Coal (Lignite), Tamil Nadu
13. Ramagundam — Andhra Pradesh (Now Telangana)
14. Obra — Russian Collaboration (UP)
15. Power Station Match (Kothagudem, Raichur, Mettur, Wankbori)
16. Paras — Maharashtra (Not Andhra Pradesh)
17. Thermal Capacity Ranking (Maharashtra > Gujarat > UP > WB)
18. Plant Load Factor (PLF) — 70-75% Average
19. Bokaro Thermal Plant — Jharkhand
20. Jajpur Steel Plant — Tata Steel (Not SAIL)
21. Barh Super Thermal Power Station — Bihar
22. Korba — Thermal (Not Wind) — Final Match Trap

### 1. Singrauli NTPC — UP-MP Border, NOT Chhattisgarh

- Singrauli mein NTPC unit 1977 mein establish hui thi, 1982 se operational hai — India ki 'oldest/first NTPC unit' hai — Chhattisgarh mein NAHI hai, balki Sidhi (MP) aur Sonbhadra (UP) districts mein failii hai (2 states, 2 districts). 3 super thermal projects yahan hain. [Q1]

### 2. Idukki — Hydro (Not Thermal), Match-Set Verification

- Idukki (Kerala, Periyar river) hydroelectric project hai, thermal NAHI — Sabarigiri-Hydro, Ghatprabha-Irrigation, Ramganga-Multipurpose sahi pairs hain. [Q2]

### 3. Thermal Plant-State Match (Ukai, Patratu, Pench, Dabhol)

- Match: Ukai — Gujarat, Patratu — Jharkhand, Pench — Madhya Pradesh, Dabhol — Maharashtra. [Q3]

### 4. Uran — Gas-Based Plant in Maharashtra

- Uran (Uran Gas Turbine Power Station, Raigad district, Maharashtra) gas-based thermal power plant hai. [Q4]

## 5. Coal — India's Largest Energy Source (52.89%)

- 31 Dec 2024 tak **Thermal energy ka sabse zyada share (52.89%, 244376.91 MW) hai** — coal 212349.50 MW, lignite 6620 MW, gas 24818.21 MW, diesel 589.20 MW contribute karta hai. [Q5]

## 6. Thermal Power — Highest Share in Generation

- Power generation mein **Thermal power ka sabse zyada share hai** — Atomic/Hydro se zyada. [Q6]

## 7. Energy Source Ranking (Disputed — No Fixed Sequence)

- 31 Dec 2024 ke data ke mutabik: **Thermal (244376.91 MW) > Wind (48163.16 MW) > Hydro (46968.17 MW) > Nuclear (8180 MW)** — diye gaye options mein koi bhi is sequence se match nahi karta, isliye ye disputed question hai. [Q7]

## 8. 2019 Installed Capacity Order (Water > Gas > Nuclear > Diesel)

- 30 Sept 2019 ke data ke mutabik descending order: **Water > Natural Gas > Nuclear > Diesel**. [Q8]

## 9. Farakka — NTPC Superthermal Plant in West Bengal

- NTPC ka superthermal plant West Bengal mein **Farakka (Murshidabad district, Nabarun)** mein hai — coal-based plant hai. [Q9]

## 10. Map-Based — Srisailem is Hydro, Not Thermal

- Map-labeling: **Ukai (Gujarat), Korba (Chhattisgarh), Obra (UP)** — thermal power stations hain; **Srisailem (Telangana)** — hydroelectric hai, thermal NAHI. [Q10]

## 11. Coal-Based Plant Match (Kahalgau-Bihar, Khargone-MP)

- Match: **Kahalgau — Bihar, Khargone — Madhya Pradesh, Korba — Chhattisgarh, Kudgi — Karnataka**. [Q11]

## 12. Neyveli — Tertiary Coal (Lignite), Tamil Nadu

- Neyveli thermal power plant (Tamil Nadu) **Tertiary coal (Lignite)** se feed hoti hai — South-East Asia ka pehla lignite-fired thermal power station hai. [Q12]

## 13. Ramagundam — Andhra Pradesh (Now Telangana)

- Ramagundam Super Thermal Power Station **Andhra Pradesh (ab Telangana)** mein hai — capacity 2600 MW. [Q13]

## 14. Obra — Russian Collaboration (UP)

- Obra Thermal Power Station (Sonbhadra, UP) **Russia ki madad se 1967 mein establish hui** — Russian engineers involved the. [Q14]

## 15. Power Station Match (Kothagudem, Raichur, Mettur, Wankbori)

- Match: **Kothagudem — Andhra Pradesh (ab Telangana), Raichur — Karnataka, Mettur — Tamil Nadu, Wankbori — Gujarat.** [Q15]

## 16. Paras — Maharashtra (Not Andhra Pradesh)

- Badarpur-Delhi, Harduaganj-UP, Utaran-Gujarat sahi hain — '**Paras-Andhra Pradesh**' **GALAT** hai (Paras asal mein Maharashtra mein hai). [Q16]

**⚠ EXAM TRAP — Paras — Maharashtra mein hai, Andhra Pradesh mein NAHI**

## 17. Thermal Capacity Ranking (Maharashtra > Gujarat > UP > WB)

- 31 Dec 2024 ki installed thermal capacity ranking: **Maharashtra (28917.02 MW) > Gujarat (25145.93 MW) > Uttar Pradesh (23673.02 MW) > West Bengal (8763.34 MW)** — diye gaye options ke sath match nahi karta, isliye disputed question hai. [Q17]

## 18. Plant Load Factor (PLF) — 70-75% Average

- All-India Thermal Plant Load Factor **70-75% ke beech vary karta hai** — Central (~71%), State (~60%), Private IPP (~65%), Private UTL (~54%) sector-wise. [Q18]

## 19. Bokaro Thermal Plant — Jharkhand

- Bokaro Thermal Power Station **Jharkhand (Bokaro district)** mein hai. [Q19]

## 20. Jajpur Steel Plant — Tata Steel (Not SAIL)

- Kalinganagar Industrial Complex (Jajpur, Odisha) ka integrated steel plant **Tata Steel ke tahat kaam karta hai — SAIL ke tahat NAHI** (question ka official answer key mismatch hai, note isi wajah se diya gaya hai). [Q20]

## 21. Barh Super Thermal Power Station — Bihar

- Barh Super Thermal Power Station **Patna district, Bihar** mein hai — total approved capacity 3300 MW, installed 1320 MW. [Q21]

## 22. Korba — Thermal (Not Wind) — Final Match Trap

- Patratu-Thermal, Jhakri-Hydroelectric, Kalpakkam-Nuclear sahi hain — '**Korba-Wind**' **GALAT** hai (Korba asal mein Thermal power station hai). [Q22]